

Generation Interconnection System Impact Study Report

For Queue Project Q19-1266

Vacherie 138 kV

July 18, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1266. The Interconnection Customer proposes a 200.2 MW solar photovoltaic facility interconnecting at the Vacherie 138 kV substation in the Virginia area. The request was received on August 08, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 334 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,477,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Mcadenville to Stock Island Substation 138 KV

Contingency: Vacherie to Friendship 138 KV line

Loading: 116.6% (Rating: 253 MVA)

DFAX: 75.4%

Pre-project loading: 56.8%

2. Facility: Mcadenville to Vachrie 138 KV

Contingency: Vacherie to Friendship 138 KV

Loading: 113.7% (Rating: 319 MVA)

DFAX: 100.0%

Pre-project loading: 51.0%

3. Facility: Vacherie to Friendship 138 KV

Contingency: Mcadenville to Vacherie 138 KV

Loading: 105.4% (Rating: 344 MVA)

DFAX: 100.0%

Pre-project loading: 47.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcadeville Substation to Stock Island 138 KV	Rebuild line with higher-capacity conductor	\$16,779,000	48% (\$8,000,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Vacherie to Mcadenville 138 KV L	Rebuild line with higher-capacity conductor	\$40,444,000	52% (\$20,966,000)
Vacherie to Friendship 138 KV	Reconductor line	\$2,914,000	52% (\$1,511,000)
	Total allocated to project		\$30,477,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.